РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №13

дисциплина: Администрирование сетевых подсистем

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МОСКВА

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Цель работы:

Целью данной работы является приобретение навыков настройки сервера NFS для удалённого доступа к ресурсам.

Выполнение работы:

```
[vagrant@server ~]$ dnf -y install nfs-utils
Error: This command has to be run with superuser privileges (under the root user on most systems).
[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install nfs-utils
Last metadata expiration check: 1:21:33 ago on Thu 12 Mar 2026 09:05:56 AM UTC.
Dependencies resolved.
================================================================================
 Package                Arch        Version                   Repository   Size
================================================================================
Installing:
 nfs-utils              x86_64      1:2.5.4-38.el9_7.3        baseos      434 k
Upgrading:
 libipa_hbac            x86_64      2.9.7-4.el9_7.1           baseos       33 k
 libldb                 x86_64      4.22.4-12.el9_7           baseos      180 k
 libsmbclient           x86_64      4.22.4-12.el9_7           baseos       73 k
 libsss_certmap         x86_64      2.9.7-4.el9_7.1           baseos       88 k
 libsss_idmap           x86_64      2.9.7-4.el9_7.1           baseos       39 k
 libsss_nss_idmap       x86_64      2.9.7-4.el9_7.1           baseos       43 k
 libsss_sudo            x86_64      2.9.7-4.el9_7.1           baseos       33 k
 libtalloc              x86_64      2.4.3-1.el9               baseos       33 k
 libtdb                 x86_64      1.4.13-1.el9              baseos       53 k
 libtevent              x86_64      0.16.2-1.el9              baseos       50 k
 libwbclient            x86_64      4.22.4-12.el9_7           baseos       42 k
 samba-client-libs      x86_64      4.22.4-12.el9_7           baseos      5.3 M
 samba-common           noarch      4.22.4-12.el9_7           baseos      173 k
 samba-common-libs      x86_64      4.22.4-12.el9_7           baseos      104 k
 sssd                   x86_64      2.9.7-4.el9_7.1           baseos       25 k
 sssd-ad                x86_64      2.9.7-4.el9_7.1           baseos      217 k
 sssd-client            x86_64      2.9.7-4.el9_7.1           baseos      158 k
 sssd-common            x86_64      2.9.7-4.el9_7.1           baseos      1.6 M
 sssd-common-pac        x86_64      2.9.7-4.el9_7.1           baseos       94 k
 sssd-ipa               x86_64      2.9.7-4.el9_7.1           baseos      286 k
 sssd-kcm               x86_64      2.9.7-4.el9_7.1           baseos      107 k
 sssd-krb5              x86_64      2.9.7-4.el9_7.1           baseos       70 k
 sssd-krb5-common       x86_64      2.9.7-4.el9_7.1           baseos       92 k
 sssd-ldap              x86_64      2.9.7-4.el9_7.1           baseos      159 k
 sssd-proxy             x86_64      2.9.7-4.el9_7.1           baseos       70 k
```

```
Open ▾   *exports
         /etc                                                    Save
1 /srv/nfs *(ro)
```

```
(gedit:7260): dconf-WARNING **: 10:35:53.460: failed to commit changes to dconf: Failed to execute
 child process “dbus-launch” (No such file or directory)
[root@server etc]# ^C
[root@server etc]#
[root@server etc]#
[root@server etc]# semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
[root@server etc]# restorecon -vR /srv/nfs
Relabeled /srv/nfs from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:nfs_t:s0
[root@server etc]# systemctl start nfs-server.service
[root@server etc]# systemctl enable nfs-server.service
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/
system/nfs-server.service.
[root@server etc]# irewall-cmd --add-service=nfs
bash: irewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=nfs
success
[root@server etc]# firewall-cmd --add-service=nfs --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]#
```

```
root@server:/etc
[root@server etc]# sof | grep TCP
bash: sof: command not found...
[root@server etc]# lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
      Output information may be incomplete.
systemd      1                         root   62u     IPv4              46424       0t0        TC
P *:sunrpc (LISTEN)
systemd      1                         root   64u     IPv6              45670       0t0        TC
P *:sunrpc (LISTEN)
cupsd      953                         root    6u     IPv6              21155       0t0        TC
P localhost:ipp (LISTEN)
cupsd      953                         root    7u     IPv4              21156       0t0        TC
P localhost:ipp (LISTEN)
sshd       968                         root    3u     IPv4              21197       0t0        TC
P *:down (LISTEN)
sshd       968                         root    4u     IPv6              21199       0t0        TC
P *:down (LISTEN)
sshd       968                         root    5u     IPv4              21201       0t0        TC
P *:ssh (LISTEN)
sshd       968                         root    6u     IPv6              21203       0t0        TC
P *:ssh (LISTEN)
mariadbd  1060                        mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1149 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1215 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1233 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1238 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1349 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
mariadbd  1060 1352 mariadbd          mysql   20u     IPv6              23706       0t0        TC
P *:mysql (LISTEN)
```

```
root@server:/etc
[root@server etc]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
      Output information may be incomplete.
systemd      1                          root   63u     IPv4              45663       0t0        UD
P *:sunrpc
systemd      1                          root   65u     IPv6              45677       0t0        UD
P *:sunrpc
avahi-dae  637                         avahi   12u     IPv4              19843       0t0        UD
P *:mdns
avahi-dae  637                         avahi   13u     IPv6              19844       0t0        UD
P *:mdns
avahi-dae  637                         avahi   14u     IPv4              19845       0t0        UD
P *:60877
avahi-dae  637                         avahi   15u     IPv6              19846       0t0        UD
P *:52703
chronyd    658                        chrony    5u     IPv4              19840       0t0        UD
P localhost:323
chronyd    658                        chrony    6u     IPv6              19841       0t0        UD
P localhost:323
chronyd    658                        chrony    7u     IPv4              19842       0t0        UD
P *:ntp
NetworkMa  928                          root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
NetworkMa  928  931 gmain               root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
NetworkMa  928  932 gdbus               root   30u     IPv4              22111       0t0        UD
P server.user.net:bootpc->_gateway:bootps
rpcbind   7310                           rpc    5u     IPv4              45663       0t0        UD
P *:sunrpc
rpcbind   7310                           rpc    7u     IPv6              45677       0t0        UD
P *:sunrpc
rpc.statd 7313                       rpcuser    7u     IPv4              54670       0t0        UD
P *:50942
rpc.statd 7313                       rpcuser    9u     IPv4              54664       0t0        UD
P localhost:708
```

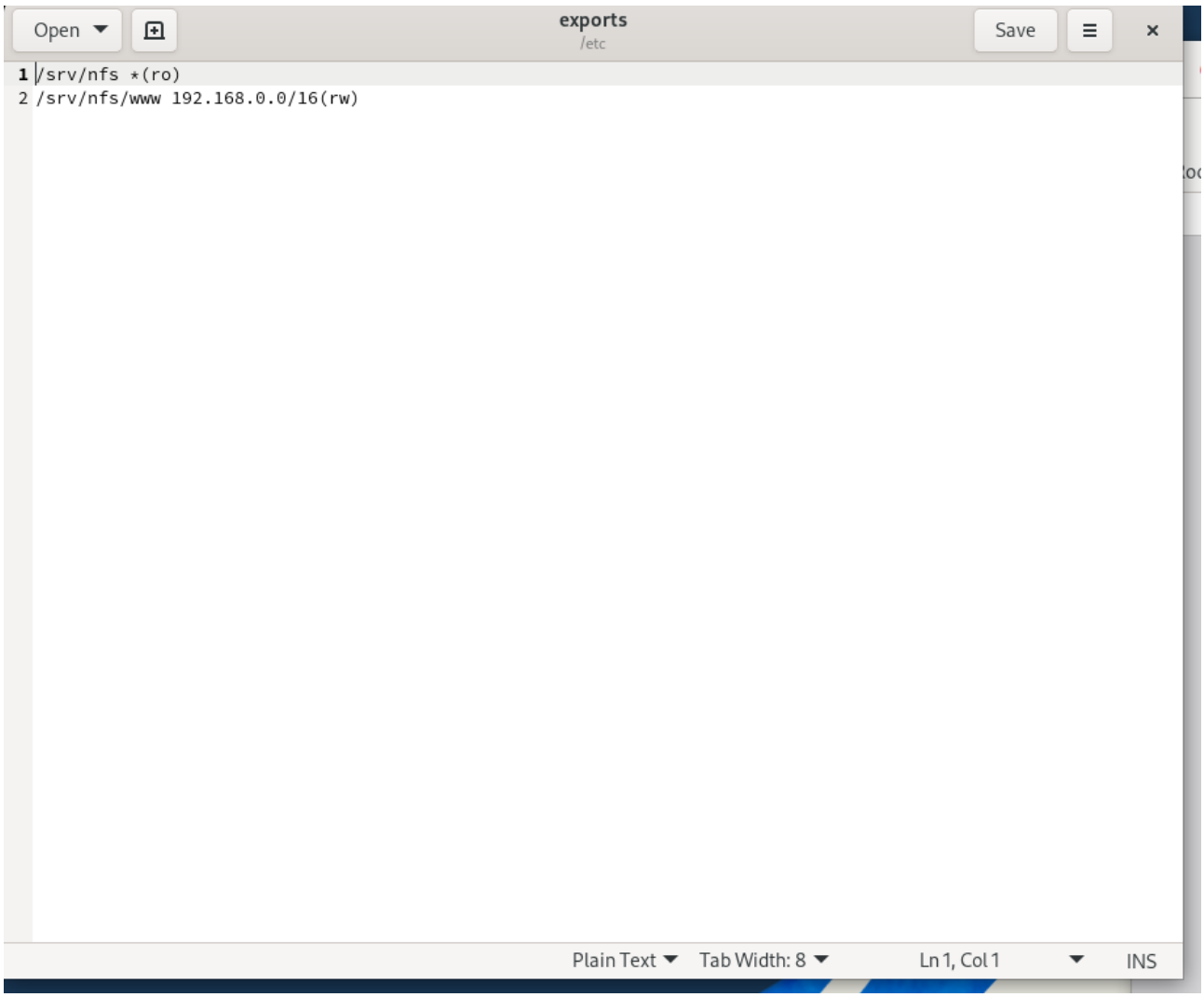
```
[root@server etc]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit
ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin b
itcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine
 checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp d
hcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticse
arch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps fre
eipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-
availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isn
s jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver ku
be-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure
kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly k
ubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-
udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmu
r mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imag
eio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresq
l privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetma
ster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba sa
mba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptra
p spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syn
cthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks tr
ansmission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discove
ry ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server etc]# irewall-cmd --add-service=mountd --add-service=rpc-bind
bash: irewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]#
```

```
[root@client ~]# mount server.user.net:/srv/nfs /mnt/nfs
mount: /mnt/nfs: bad option; for several filesystems (e.g. nfs, cifs) you might need a /sbin/mount
.<type> helper program.
[root@client ~]# mount
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4096k,nr_inodes=448710,mode=755,inode64)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=729744k,nr_inodes=819200,mode=755,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory
_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
/dev/sda1 on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=29,pgrp=1,timeout=0,minproto=5,m
axproto=5,direct,pipe_ino=19574)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,relatime,seclabel,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-sysctl.service type ramfs (ro,nosuid,nodev,noexec,relatime,seclab
el,mode=700)
```

```
p_id=1000)
[root@client ~]# suytemctl status remote-fs.target
bash: suytemctl: command not found...
[root@client ~]# systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
     Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset: enabled)
     Active: active since Thu 2026-03-12 10:45:13 UTC; 52min ago
      Until: Thu 2026-03-12 10:45:13 UTC; 52min ago
       Docs: man:systemd.special(7)

Mar 12 10:45:13 client.user.net systemd[1]: Reached target Remote File Systems.
[root@client ~]#
```

```
l privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetma
ster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba sa
mba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptra
p spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syn
cthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks tr
ansmission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discove
ry ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server etc]# irewall-cmd --add-service=mountd --add-service=rpc-bind
bash: irewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]# mkdir -p /srv/nfs/www
[root@server etc]# mount -o bind /var/www/ /srv/nfs/www/
[root@server etc]#
```


Open
exports
/etc
Save
/srv/nfs *(ro)
/srv/nfs/www 192.168.0.0/16(rw)
Plain Text
Tab Width: 8
Ln 1, Col 1
INS

Open Save

*fstab
/etc

```
1
2 #
3 # /etc/fstab
4 # Created by anaconda on Fri Jul 26 13:25:41 2024
5 #
6 # Accessible filesystems, by reference, are maintained under '/dev/disk/'.
7 # See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
8 #
9 # After editing this file, run 'systemctl daemon-reload' to update systemd
10 # units generated from this file.
11 #
12 UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                       xfs     defaults        0 0
13 /swapfile none swap defaults 0 0
14 #VAGRANT-BEGIN
15 # The contents below are automatically generated by Vagrant. Do not modify.
16 #VAGRANT-END
17
18 /var/www /srv/nfs/www none bind 0 0
```

```
(gedit:7666): dconf-WARNING **: 12:08:58.809: failed to commit changes to dconf: Failed to execute
 child process “dbus-launch” (No such file or directory)
[root@server etc]# exportfs -r
[root@server etc]# mkdir -p -m 700 ~/common
[root@server etc]# cd ~/common
[root@server common]# touch user@server.txt
[root@server common]# mkdir -p /srv/nfs/home/user
[root@server common]# mkdir -p /srv/nfs/home/vagrant
[root@server common]# mount -o bind /home/user/common /srv/nfs/home/vagrant
mount: /srv/nfs/home/vagrant: special device /home/user/common does not exist.
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
[root@server common]#
```

Open ▾ | exports /etc | Save

```
/srv/nfs *(ro)
/srv/nfs/www 192.168.0.0/16(rw)
/srv/nfs/home/user 192.168.0.0/16(rw)
```

Open ▾ | fstab /etc | Save

```

#
# /etc/fstab
# Created by anaconda on Fri Jul 26 13:25:41 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                       xfs     defaults        0 0
/swapfile none swap defaults 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
#VAGRANT-END

/var/www /srv/nfs/www none bind 0 0
/home/user/common /srv/nfs/home/vagrant none bind 0 0
```

Saving file "/etc/fstab"... | Plain Text ▾ | Tab Width: 8 ▾ | Ln 19, Col 54 ▾ | INS

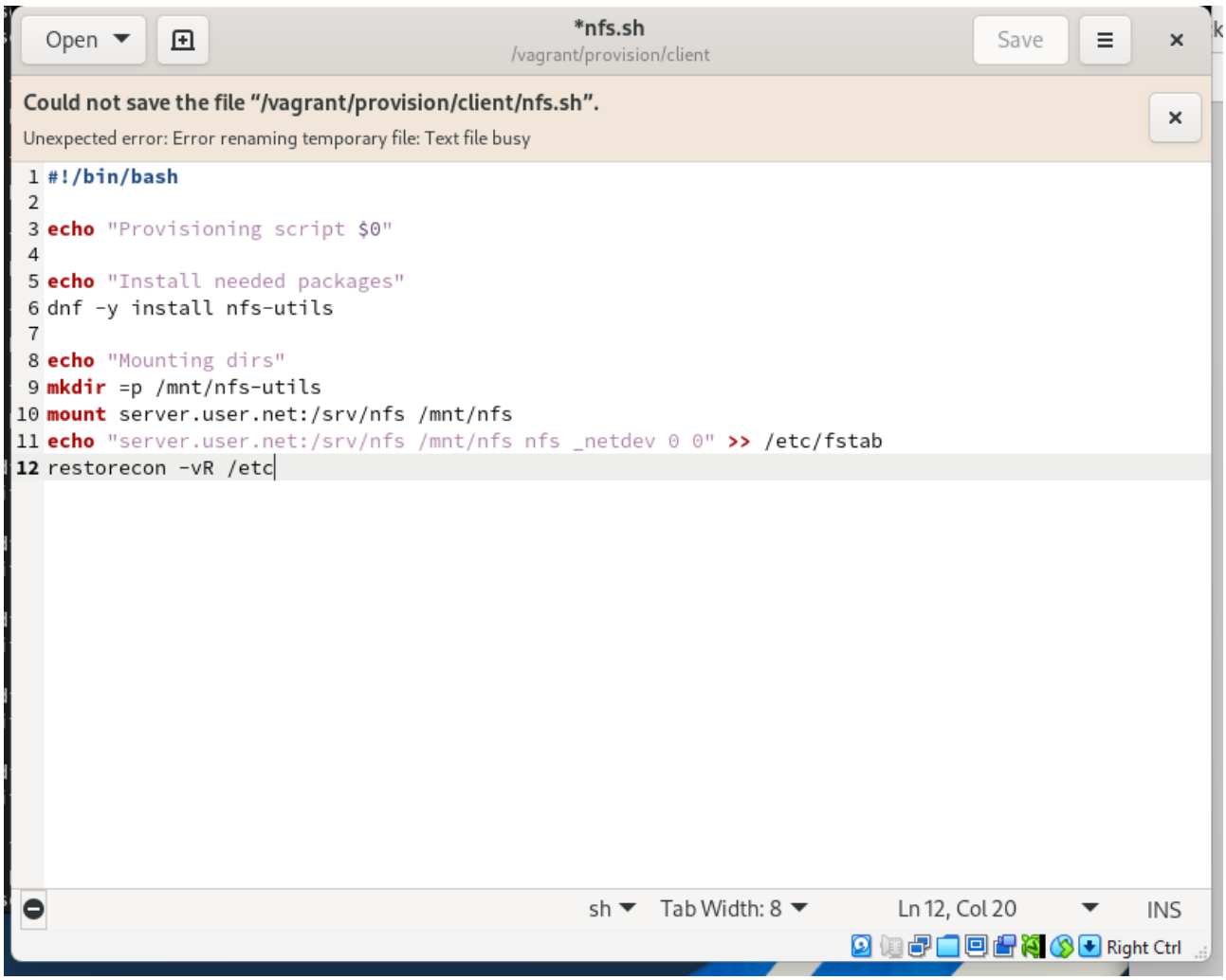
```
[root@client ~]# suytemctl status remote-fs.target
bash: suytemctl: command not found...
[root@client ~]# systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
     Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset: enabled)
     Active: active since Thu 2026-03-12 10:45:13 UTC; 52min ago
      Until: Thu 2026-03-12 10:45:13 UTC; 52min ago
       Docs: man:systemd.special(7)

Mar 12 10:45:13 client.user.net systemd[1]: Reached target Remote File Systems.
[root@client ~]# cd /mnt/nfs/home/user
-bash: cd: /mnt/nfs/home/user: No such file or directory
[root@client ~]# cd /mnt/nfs/home
-bash: cd: /mnt/nfs/home: No such file or directory
[root@client ~]# mkdir  /mnt/nfs/home/user
mkdir: cannot create directory '/mnt/nfs/home/user': No such file or directory
[root@client ~]# mkdir -p /mnt/nfs/home/user
[root@client ~]# cd /mnt/nfs/home/user
[root@client user]# touch user@client.txt
[root@client user]#
```

Open ▾ | nfs.sh — /vagrant/provision/server | Save

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install nfs-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/nfs/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service nfs --permanent
firewall-cmd --add-service mountd --add-service rpc-bind --permanent
firewall-cmd --reload
echo "Tuning SELinux"
mkdir -p /srv/nfs
semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
restorecon -vR /srv/nfs
echo "Mounting dirs"
mkdir -p /srv/nfs/www
mount -o bind /var/www /srv/nfs/www
echo "/var/www /srv/nfs/www none bind 0 0" >> /etc/fstab
mkdir -p /srv/nfs/home/user
mkdir -p -m 700 /home/user/common
chown user:user /home/user/common
mount -o bind /home/user/common /srv/nfs/home/user
echo "/home/user/common /srv/nfs/home/user none bind 0 0" >> /etc/fstab
echo "Start nfs service"
systemctl enable nfs-server
systemctl start nfs-server
systemctl restart firewalld
```

sh ▾ | Tab Width: 8 ▾ | Ln 19, Col 57 | INS


Open
*nfs.sh
/vagrant/provision/client
Save
Could not save the file "/vagrant/provision/client/nfs.sh".
Unexpected error: Error renaming temporary file: Text file busy
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install nfs-utils
echo "Mounting dirs"
mkdir =p /mnt/nfs-utils
mount server.user.net:/srv/nfs /mnt/nfs
echo "server.user.net:/srv/nfs /mnt/nfs nfs _netdev 0 0" >> /etc/fstab
restorecon -vR /etc
sh
Tab Width: 8
Ln 12, Col 20
INS
Right Ctrl

Вывод:

В ходе выполнения лабораторной работы были приобретены навыки настройки сервера NFS для удалённого доступа к ресурсам.

Ответы на контрольные вопросы:

1. Как называется файл конфигурации, содержащий общие ресурсы NFS? - **Файл конфигурации, содержащий общие ресурсы NFS, называется /etc/exports. В этом файле определяются каталоги, которые будут доступны для общего использования через NFS.**
2. Какие порты должны быть открыты в брандмауэре, чтобы обеспечить полный доступ к серверу NFS? **- Для обеспечения полного доступа к серверу NFS, обычно открываются следующие порты:**

   TCP и UDP порт 2049: Основной порт для NFS.

   TCP и UDP порт 111: Порт для службы rpcbind (или portmap), которая используется для регистрации служб RPC.

   Порты для динамически выделяемых портов (обычно в диапазоне 32768-32779), используемых NFS для передачи данных.

3. Какую опцию следует использовать в /etc/fstab, чтобы убедиться, что общие ресурсы NFS могут быть установлены автоматически при перезагрузке? **- Для автоматической установки общих ресурсов NFS при загрузке системы, в файле /etc/fstab следует использовать опцию auto. Пример строки в /etc/fstab:**

   server:/remote/export /mnt/nfs_share nfs auto 0 0

   Здесь server:/remote/export - адрес и путь к общему ресурсу NFS, /mnt/nfs_share - точка монтирования, nfs - тип файловой системы, auto - опция для автоматического монтирования при загрузке

системы, 0 0 - опции для процесса проверки файловой системы при загрузке.